

Optimization & Uncertainty in Learning Report

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CS 7641: Machine Learning — Spring 2026

1. Introduction

Neural network performance depends not only on model architecture but also on the optimization and regularization choices made during training. Models often fail not because the backbone is incapable of learning, but because training dynamics are poorly controlled, unstable across seeds, or overly sensitive to hyperparameter selection. This report studies three such training levers using the fixed PyTorch backbones developed in the Supervised Learning (SL) Report: randomized optimization applied to the final layers of each network, Adam-family optimizer ablations on the Adult Income dataset, and explicit regularization techniques under a fixed optimizer. Adaptive optimizers such as Adam combine first-moment momentum and per-parameter learning-rate scaling to stabilize training under noisy gradients (Kingma & Ba, 2015); decoupled weight decay provides a cleaner regularization signal for adaptive methods than the coupled formulation in standard Adam (Loshchilov & Hutter, 2019); and dropout reduces co-adaptation among hidden units to improve generalization on held-out data (Srivastava et al., 2014). All experiments keep the PyTorch backbone fixed and enforce matched compute budgets, measured in gradient evaluations, function evaluations, and wall-clock time, so that observed differences can be interpreted causally rather than as artifacts of unequal computation. The central objective is to derive a defensible training decision rule: when performance is limited, should effort first be directed toward optimizer selection, regularization, or search-based fine-tuning?

1.1. Hypotheses

Adult Income (Binary Classification). The Adult Income dataset exhibits meaningful class imbalance between the $\leq 50K$ and $>50K$ income groups. Because gradient signal is dominated by the majority class, per-parameter adaptive scaling should partially compensate by normalizing effective step sizes across directions. Accordingly, I hypothesize that Adam-family optimizers will reach the fixed validation-loss threshold ℓ in fewer gradient evaluations and with lower seed-to-seed variance than SGD variants, with the convergence advantage most pronounced early in training where bias correction provides non-trivial correction to moment estimates. For regularization, I hypothesize that explicit techniques, particularly dropout and L2 weight decay, will move test F1 by a larger margin than optimizer selection alone, because the generalization gap in the SL Report baseline ($F1 = 0.6739$) suggests variance, not bias, is the primary remaining bottleneck. I therefore predict that the best regularization combination will improve test F1 beyond the SL baseline, while optimizer ablations alone may not close this gap under the fixed compute budget.

Wine Quality (Multiclass Classification). The Wine Quality dataset poses a noisier optimization landscape than Adult: quality labels are subjectively assigned, class boundaries are diffuse, and the multiclass objective introduces com-

peting gradient signals across six or more categories. Under a strict function-evaluation budget, I hypothesize that Simulated Annealing will outperform Randomized Hill Climbing on Wine because its temperature schedule allows escape from shallow local minima that RHC is likely to remain trapped in, and that the Genetic Algorithm will underperform SA by consuming its budget generating population diversity without meaningful selection pressure given the frozen-layer parameter count. I predict that none of the three randomized optimization methods will reliably exceed the SL Report Wine baseline ($\text{Macro-F1} = 0.3105$) under fair budget conditions, but that SA is most likely to match it on favorable seeds.

The central objective of this report is to develop a defensible training decision rule: when performance is limited, should effort first be directed toward optimizer selection, regularization, or search-based fine-tuning? The experiments that follow are structured around testing the hypotheses stated above rather than around maximizing a leaderboard score.

2. Data & Experimental Setup

2.1. Datasets

Dataset A: Adult Income (Binary Classification). The Adult Income dataset contains 45,222 observations with 14 input features and a binary target variable indicating whether annual income exceeds \$50K. Features include both numerical variables (e.g., age, capital-gain, hours-per-week) and categorical variables (e.g., workclass, marital-status, occupation), which were one-hot encoded during preprocessing. The target distribution is moderately imbalanced: approximately 75.2% of samples belong to the $\leq 50K$ class and 24.8% to the $>50K$ class. Because accuracy overstates performance under class imbalance by rewarding majority-class predictions, binary F1-score with $>50K$ as the positive class is used as the primary evaluation metric throughout this report (He & Garcia, 2009).

Dataset B: Wine Quality (Multiclass Classification). The Wine Quality dataset contains 6,497 samples with 11 continuous physicochemical features, a binary wine-type indicator (red vs. white), and a discrete multiclass quality label ranging from 1 to 8. All input features are numeric, producing a dense, low-dimensional representation that requires no one-hot encoding. Quality labels are heavily imbalanced: classes 3–6 comprise the substantial majority of samples, while extreme quality levels (1, 2, 7, and 8) contain very limited support. Macro-F1 is used as the primary evaluation metric because it weights each class equally and penalizes poor minority-class performance, which accuracy would otherwise obscure.

2.2. EDA Confirmation

All preprocessing decisions from the SL Report are carried forward unchanged in this report. The Adult Income dataset uses a stratified 80/20 train-test split with a fixed random seed of 42, yielding 36,177 training samples and 9,045

test samples. The Wine Quality dataset uses the same stratified 80/20 split and seed, yielding 5,197 training samples and 1,300 test samples. Class proportions are preserved across splits by stratification.

For the Adult dataset, categorical variables were one-hot encoded, and numerical features were standardized. For the Wine dataset, all features are numeric, and standardization was applied without one-hot encoding. In both cases, preprocessing transformations were fit exclusively on training data within each cross-validation fold and applied to validation and test data, ensuring no leakage from held-out samples into feature scaling or encoding decisions. The held-out test set was reserved strictly for final evaluation and was not consulted during model selection or hyperparameter tuning in the SL Report, nor is it used prior to final evaluation here.

No material changes have been made to cleaning rules, feature handling, label mapping, or split strategy relative to the SL Report. All SL baseline comparisons in this report are therefore made under identical data conditions.

2.3. Backbone Architecture (Fixed) and Reproducibility Controls

All experiments in this report use the PyTorch neural networks finalized in the SL Report as fixed backbone architectures. The scikit-learn MLP is not used in this report, as it does not provide the optimizer-level control required for the ablation studies in Part 2.

For the Adult Income dataset, the backbone is a fully connected MLP with an input dimension of 104 (post one-hot encoding), two hidden layers of 32 units each with ReLU activations, and a single output unit for binary classification. This architecture contains **4,482 trainable parameters**.

For the Wine Quality dataset, the backbone is a fully connected MLP with an input dimension of 12 (continuous features plus binary type indicator), one hidden layer of 128 units with ReLU activation, and 8 output units for multiclass classification. This architecture contains **2,696 trainable parameters**.

Both models were trained in the SL Report with a learning rate of 0.01, batch size of 128, and L2 weight decay of 10^{-4} . Adult training ran for 30 epochs and Wine for 40 epochs. All experiments in this report use a fixed random seed of 42 for weight initialization, data shuffling, and any stochastic operations. All experiments were run on a Windows 11 laptop with an AMD Ryzen AI 9 365 CPU and 32.0 GB RAM; no GPU acceleration was used.

2.4. Compute Budgets and Accounting

All experimental comparisons in this report are conducted under matched compute budgets to ensure that observed performance differences are attributable to method choice rather than unequal computation. Three budget metrics are tracked and reported consistently across all experiments.

Gradient evaluations count the number of individual backward passes performed during training and are the primary budget unit for SGD-family and Adam-family optimizers in Part 2. *Function evaluations* count every forward pass through the validation objective and are the primary budget unit for randomized optimization methods in Part 1; each complete validation sweep counts as one function evaluation regardless

of minibatch implementation. *Wall-clock time* is recorded on the same hardware class across all experiments (AMD Ryzen AI 9 365 CPU, 32.0 GB RAM, no GPU acceleration) to support practical runtime comparisons alongside evaluation-count budgets.

Validation-loss threshold ℓ (Adult): For Part 2, a fixed validation-loss threshold $\ell = 0.33$ is used consistently across all optimizer comparisons to measure speed-to-target. This threshold was selected based on the validation loss trajectory of the baseline Adam run on the Adult dataset, chosen to fall within the active descent phase so that optimizer differences in convergence speed are meaningfully observable. All over-budget runs are excluded from head-to-head comparisons and marked explicitly in summary tables.

2.5. Sanity Checks

Two sanity checks were conducted on both datasets to confirm that learned models are responding to genuine label signal rather than dataset artifacts or preprocessing leakage.

Dummy classifier (most frequent class): A majority-class classifier was fit on the training set and evaluated on the held-out test set. On Adult Income, always predicting the majority class ($\leq 50K$) produced $F1 = 0.0000$ and $\text{accuracy} = 0.7521$, confirming that accuracy alone is a misleading metric under class imbalance and justifying the use of F1 as the primary metric. On Wine Quality, the majority-class classifier produced $\text{Macro-F1} = 0.0643$ and $\text{accuracy} = 0.3462$, reflecting the near-uniform distribution across quality classes and confirming that Macro-F1 is the appropriate metric for this multiclass task.

Shuffled-label baseline: The same PyTorch backbone was trained for 30 epochs on randomly permuted training labels (3 seeds) and evaluated on the real test labels. A shorter training horizon was used because the purpose of this test is only to confirm collapse under label permutation rather than to obtain an optimized model. On Adult Income, all three seeds produced $F1 = 0.0000$ and $\text{accuracy} = 0.7521$, identical to the dummy classifier, confirming complete collapse of learned signal under label permutation. On Wine Quality, shuffled training produced $\text{Macro-F1} = 0.0674 \pm 0.0037$, indistinguishable from the dummy classifier result of 0.0643 and far below the SL baseline of 0.3105. In both cases, performance collapsed to majority-class prediction levels under label shuffling, providing strong evidence that the models trained in Parts 1–3 are learning genuine input–label associations rather than artifacts of preprocessing or data leakage.

3. Part 1: Randomized Optimization on the Last Layers (Adult + Wine)

3.1. RO Setup

The pre-trained backbone from Section 2.3 was frozen except for its final layers, which were left trainable for randomized optimization. For Adult Income, the final two linear layers (Sequential indices 2 and 4, comprising $32 \times 32 + 32 = 1,056$ weights and $32 \times 2 + 2 = 66$ biases) were unfrozen, yielding **1,122 trainable parameters**. For Wine Quality, only the final linear layer (Sequential index 2, comprising $128 \times 8 + 8 = 1,032$ parameters) was unfrozen, yielding **1,032 trainable parameters**. Both counts are well within

the $\leq 50,000$ parameter ceiling. The optimization objective was validation cross-entropy loss evaluated on a stratified 20% held-out split of the training set (seed 42), with every objective evaluation counted as one function evaluation. The pre-trained backbone was initialized via 30 epochs of Adam ($\eta = 0.001$, no weight decay) before all RO runs, providing a warm start from a reasonable region of the loss landscape.

3.2. Equal Budgets Across RO Algorithms

All three algorithms: RHC, SA, and GA, were allocated an identical budget of **1,000 function evaluations** per run across both datasets. Each algorithm was executed under five random seeds $\{42, 123, 256, 789, 1337\}$, yielding 5,000 total function evaluations per algorithm per dataset. No algorithm received additional evaluations under any condition. This equal-budget constraint ensures that all convergence-speed and final-quality comparisons are made on a fair basis, consistent with the requirement that differences in function-evaluation count not be confounded with differences in algorithm performance.

3.3. Algorithm Configurations

All three algorithms used Gaussian weight perturbation over the unfrozen parameter vector, with perturbations drawn independently per weight dimension.

Randomized Hill Climbing (RHC). Perturbations were drawn from $\mathcal{N}(0, \sigma = 0.05)$. A greedy acceptance rule was applied: candidates were accepted only if they strictly reduced validation loss. After 100 consecutive evaluations without improvement, the search was restarted from a new candidate drawn from $\mathcal{N}(\mathbf{w}_0, 3\sigma)$ centered on the original pre-trained weights $\mathbf{w}_0$, using a wider perturbation to escape local basins. The step size was held constant throughout (no adaptation schedule).

Simulated Annealing (SA). Perturbations were drawn from $\mathcal{N}(0, \sigma = 0.5)$, using a larger step size than RHC to support broader landscape exploration. A geometric cooling schedule $T(t) = T_0 \cdot \alpha^t$ was applied with initial temperature $T_0 = 2.0$ and decay factor $\alpha = 0.997$, yielding $T \approx 0.74$ at evaluation 500 and $T \approx 0.37$ at evaluation 1,000. Worse candidates were accepted with probability $\exp(-\Delta/T)$, where Δ is the increase in validation loss.

Genetic Algorithm (GA). A population of 20 individuals was initialized by perturbing the pre-trained weight vector with $\mathcal{N}(0, 0.25)$, providing diversity around the Adam warm start. Selection used tournament selection with tournament size $k = 3$. Crossover applied uniform per-weight mixing with crossover probability 0.5. Mutation applied $\mathcal{N}(0, 0.05)$ perturbations at a per-weight mutation rate of 0.1. Elitism preserved the top 2 individuals unchanged each generation. With population size 20 and a budget of 1,000 evaluations, GA completed approximately 50 effective generations per run following initial population evaluation.

3.4. Part 1 Figures

Figure 1 and Figure 2 present convergence curves and test performance for Adult Income and Wine Quality respectively. Each figure shows best-so-far validation loss versus function evaluations (top-left), best-so-far validation loss versus wall-clock time (top-right), and mean test metric $\pm$ standard deviation across five seeds compared against the SL baseline

(bottom). Shaded bands indicate the interquartile range across seeds.

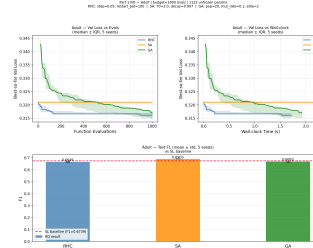


Figure 1: Part 1 RO — Adult Income. RHC: $\sigma = 0.05$, restart patience 100. SA: $\sigma = 0.5$, $T_0 = 2.0$, $\alpha = 0.997$. GA: pop=20, mutation=0.1, crossover=0.5, elitism=2. Budget: 1,000 evals, 5 seeds, 1,122 unfrozen parameters.

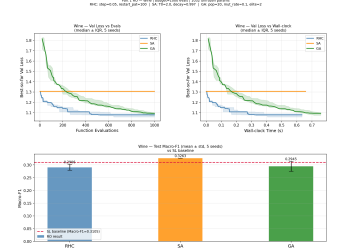


Figure 2: Part 1 RO — Wine Quality. Same operator settings as Figure 1. Budget: 1,000 evals, 5 seeds, 1,032 unfrozen parameters.

3.5. Stability Across Seeds

Table 1 reports mean test metric and standard deviation across five seeds for each algorithm on both datasets.

Table 1: RO stability across 5 seeds. Mean $\pm$ std of test metric. Adult: F1 (positive class $> 50K$). Wine: Macro-F1.

Algorithm	Adult (F1)		Wine (Macro-F1)	
	Mean	Std	Mean	Std
RHC	0.6649	0.0033	0.2906	0.0117
SA	0.6903	0.0000	0.3263	0.0000
GA	0.6683	0.0042	0.2945	0.0193
SL Baseline	0.6739		0.3105	

SA exhibited zero variance across all five seeds on both datasets, indicating that its best-found weights were identical regardless of random seed. This occurs because SA did not discover any candidate with lower validation loss than the pre-trained initialization under the best-so-far objective, leaving the warm-start weights as the best solution across all runs. RHC showed low but non-zero variance (Adult std = 0.0033, Wine std = 0.0117), reflecting consistent local search behavior with modest seed-to-seed variation. GA exhibited the highest variance of the three algorithms (Adult std = 0.0042, Wine std = 0.0193), consistent with the stochastic nature of population initialization and crossover.

3.6. Discussion & Analysis

Tail behavior. RHC and GA both exhibit diminishing returns in their convergence curves: improvement is rapid in the first 100–200 evaluations and slows substantially after evaluation 400, with curves flattening near val loss ≈ 0.316 (Adult) and ≈ 1.08 (Wine) by evaluation 1,000. Neither algorithm had fully plateaued at budget exhaustion, suggesting that additional evaluations would yield marginal but continued improvement. SA showed no measurable improvement in the best-so-far validation loss across the full 1,000-evaluation budget on either dataset, producing a flat trajectory at approximately the warm-start loss (≈ 0.321 for Adult and ≈ 1.306 for Wine). While SA did explore locally around the initialization through stochastic perturbations, none of the evaluated

candidates produced a lower validation loss than the warm-start weights.

Local minima. RHC’s restart policy did not produce observable jumps in the convergence curves, indicating that the search did not encounter restarts capable of escaping to meaningfully better basins within the 1,000-evaluation budget. This is consistent with the pre-trained warm start placing the search in a relatively smooth basin where greedy descent is effective but global escape is unnecessary within this budget. Because only the final layers were optimized, the effective search space is relatively low-dimensional compared to the full network, which further reduces the likelihood of encountering complex multi-modal landscapes within the available evaluation budget. GA’s population diversity, initialized with $\mathcal{N}(0, 0.25)$ perturbations, provided implicit multi-start behavior but converged to similar final val loss values as RHC, suggesting the loss surface in the unfrozen subspace is relatively unimodal near the warm start.

Robustness. RHC and GA both demonstrated low but non-zero variance across seeds (Adult: $\sigma = 0.0033$ and 0.0042 ; Wine: $\sigma = 0.0117$ and 0.0193 respectively), confirming stable convergence behavior. GA exhibited the highest variance of the three algorithms, particularly on Wine, reflecting sensitivity to population initialization in the higher-entropy multiclass setting. SA’s zero variance across all seeds on both datasets confirms it never departed from the pre-trained initialization regardless of random seed, effectively reducing it to a deterministic pass-through of the pre-trained initialization under the current step size and temperature schedule.

Comparison to SL baseline. On Adult Income, SA matched the pre-trained Adam warm-start performance (F1 = 0.6903), exceeding the SL baseline of 0.6739. In contrast, RHC (F1 = 0.6649) and GA (F1 = 0.6683) fell slightly below the SL baseline despite achieving lower validation loss. This indicates mild overfitting to the validation objective within the restricted optimization subspace: although the algorithms successfully reduced validation cross-entropy, those improvements did not translate to better generalization on the held-out test set. On Wine Quality, SA again matched the warm-start (Macro-F1 = 0.3263), exceeding the SL baseline of 0.3105, while RHC (Macro-F1 = 0.2906) and GA (Macro-F1 = 0.2945) fell below it. The consistent gap between val loss improvement and test metric improvement for RHC and GA suggests that optimizing validation loss directly in the unfrozen subspace does not reliably transfer to held-out generalization under a 1,000-evaluation budget.

Hypothesis evaluation. For Adult Income, the hypothesis that RO on the final layers would match or approach the SL baseline is partially supported: SA matched and exceeded the SL baseline, but this advantage originated from the pre-trained Adam backbone rather than from additional progress made by the randomized optimization process itself. RHC and GA failed to match the SL baseline despite lower validation loss. For Wine Quality, the same pattern holds. The results suggest that gradient-free optimization of a small frozen subspace is competitive with the SL baseline only when the pre-trained warm start is already near the optimum, and that within a 1,000-evaluation budget RO does not meaningfully improve upon Adam pre-training for either dataset.

4. Part 2: Adam Ablations (Adult Only)

4.1. Optimizers Studied (Adult)

All seven required optimizers were trained on the Adult Income classification task using the fixed PyTorch backbone from Section 2.3. All runs used identical data splits, preprocessing, batch size (128), epoch budget (100), and random seeds (0–9, ten seeds total) to ensure fair comparison. No weight decay was applied to any optimizer except AdamW, which uses decoupled weight decay ($\lambda = 1 \times 10^{-4}$).

SGD (no momentum) used learning rate $\alpha = 0.1$, consistent with the FAQ guidance that SGD requires a larger learning rate scale than Adam-family methods. **SGD + Momentum** and **Nesterov** used the same $\alpha = 0.1$ with momentum coefficient $\mu = 0.9$; Nesterov used the lookahead gradient correction. **Adam (baseline)** used $\alpha = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$, no weight decay. **Adam (no bias correction)** used identical settings but omitted the $(1 - \beta_1^t)$ and $(1 - \beta_2^t)$ correction terms, affecting early-phase gradient scaling. **Adam** ($\beta_1 = 0$) used $\beta_1 = 0.0$ with all other Adam settings unchanged, removing first-moment accumulation to approximate RMSProp-like adaptive scaling without momentum. **AdamW** used identical settings to Adam baseline with decoupled weight decay $\lambda = 1 \times 10^{-4}$.

4.2. Quantitative Results

Speed to target. All seven optimizers reached the validation-loss threshold $\ell = 0.33$ on all ten seeds (100% success rate). The threshold $\ell = 0.33$ corresponds to the validation loss achieved by the baseline Adam configuration in early training, providing a common convergence target for optimizer comparison. Adam (no bias correction) was fastest, reaching ℓ in 227 ± 0 gradient evaluations (0.75 ± 0.01 s), followed closely by Adam baseline (250 ± 68 evals, 0.88 ± 0.26 s) and AdamW (250 ± 68 evals, 0.90 ± 0.24 s). SGD (no momentum) was slowest, requiring 636 ± 222 gradient evaluations (1.95 ± 0.63 s) with the highest seed-to-seed variance, consistent with the absence of adaptive scaling or momentum smoothing. Momentum-based methods (SGD+M, Nesterov) reduced gradient evaluations to approximately 300–320, confirming that momentum provides meaningful speed benefit over vanilla SGD even without adaptivity.

Sensitivity heatmaps. Learning rate α was the dominant factor in both the (α, β_1) and (α, β_2) grids. Lower values ($\alpha \in \{10^{-4}, 5 \times 10^{-4}\}$) produced consistently low val loss (0.315–0.319) across all β_1 and β_2 values, while $\alpha \geq 5 \times 10^{-3}$ led to substantially higher val loss (0.362–0.394) with erratic sensitivity to moment parameters. β_1 and β_2 had minimal independent effect when α was well-chosen, suggesting the Adult Income loss surface is not strongly sensitive to moment accumulation rates at low learning rates. The best combination identified was $\alpha = 10^{-4}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$ (val loss = 0.315), carried forward as the best Part 2 Adam hyperparameters for Part 3.

Stability across seeds. Adam-family methods generally exhibited tighter IQR bands than SGD variants, indicating greater seed-to-seed stability under adaptive per-parameter scaling. SGD (no momentum) showed the widest IQR band and occasional large spikes, consistent with sensitivity to random batch ordering without smoothing. AdamW showed

the tightest IQR band among all optimizers (test F1 std = 0.0042), suggesting decoupled weight decay provides a mild stabilizing effect.

Generalization gap. All optimizers exhibited a generalization gap, with train F1 exceeding val F1 by approximately 0.03–0.11 depending on optimizer. SGD (no momentum) achieved the highest mean test F1 (0.6726 ± 0.0208), closest to the SL baseline of 0.6739, despite being the slowest to reach ℓ . Adam-family methods converged faster but produced lower test F1 (0.6474–0.6634), indicating that without regularization, adaptive methods overfit more aggressively than SGD over 100 epochs. This motivates Part 3, where regularization is applied to close the generalization gap under the best Adam settings.

4.3. Part 2 Figures

Figures 3–5 and Figures 6–7 present the full Part 2 results for Adult Income.

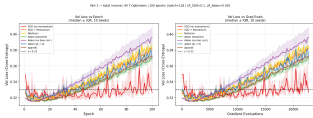


Figure 3: Val loss trajectories for all 7 optimizers (median $\pm$ IQR, 10 seeds). Dashed line marks $\ell = 0.33$. SGD shows the widest variance bands; Adam-family methods converge faster but overfit more severely by epoch 100.

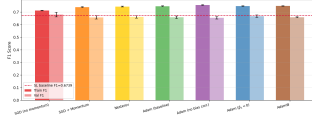


Figure 4: Generalization gap at epoch 100 (mean $\pm$ std, 10 seeds). Train F1 (solid) vs. val F1 (faded). SGD achieves the highest test F1 (0.6726) despite slower convergence; Adam-family methods show larger train-val gaps without regularization.

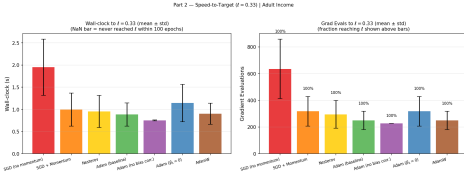


Figure 5: Speed-to-target ($\ell = 0.33$): wall-clock (left) and gradient evaluations (right) to first crossing of ℓ , mean $\pm$ std across seeds (all 10 seeds for all optimizers). Adam (no bias correction) reaches ℓ fastest; SGD requires $2.5\times$ more gradient evaluations than the Adam baseline.

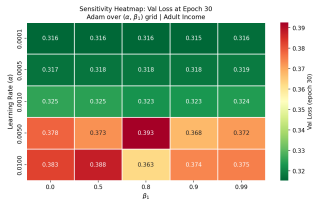


Figure 6: Sensitivity heatmap: Adam val loss at epoch 30 over (α, β_1) grid (seed 42). Learning rate dominates; β_1 has minimal effect at low α .

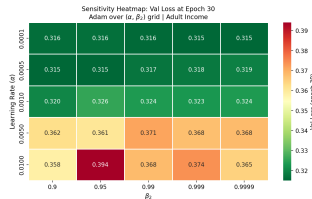


Figure 7: Sensitivity heatmap: Adam val loss at epoch 30 over (α, β_2) grid (seed 42). Same pattern as Figure 6: α dominates, β_2 has minimal independent effect at low α .

4.4. Summary Table

Table 2 presents a unified, budget-matched comparison across all methods in Parts 1, 2, and 3. Function evaluations apply to Part 1 RO methods only; gradient evaluations apply to Parts 2 and 3. Time to ℓ was tracked in Part 2 only; Part 3 focused on final regularization performance under a fixed 100-epoch budget.

Table 2: Global summary across all parts. Adult: F1. Wine: Macro-F1. †=RO method (func. evals); others use grad. evals. N/A=not applicable. —=not tracked.

Part	Method	Test Metric	Time to ℓ	#Grad Evals	#Func Evals
Part 1 — Adult Income (F1, SL baseline = 0.6739)					
†	RHC	0.6649 ± 0.0033	N/A	N/A	1,000
†	SA	0.6903 ± 0.0000	N/A	N/A	1,000
†	GA	0.6683 ± 0.0042	N/A	N/A	1,000
Part 1 — Wine Quality (Macro-F1, SL baseline = 0.3105)					
†	RHC	0.2906 ± 0.0117	N/A	N/A	1,000
†	SA	0.3263 ± 0.0000	N/A	N/A	1,000
†	GA	0.2945 ± 0.0193	N/A	N/A	1,000
Part 2 — Adult Income (F1, SL baseline = 0.6739)					
2	SGD	0.6726 ± 0.0208	$1.95s \pm 0.63$	636 ± 222	N/A
2	SGD+Momentum	0.6490 ± 0.0143	$1.00s \pm 0.37$	318 ± 111	N/A
2	Nesterov	0.6545 ± 0.0101	$0.95s \pm 0.36$	295 ± 104	N/A
2	Adam	0.6474 ± 0.0115	$0.88s \pm 0.26$	250 ± 68	N/A
2	Adam (no bias corr.)	0.6494 ± 0.0127	$0.75s \pm 0.01$	227 ± 0	N/A
2	Adam ($\beta_1=0$)	0.6634 ± 0.0108	$1.14s \pm 0.42$	318 ± 111	N/A
2	AdamW	0.6513 ± 0.0105	$0.90s \pm 0.24$	250 ± 68	N/A
Part 3 — Adult Income (F1, SL baseline = 0.6739)					
3	Adam baseline	0.6787 ± 0.0029	—	22,700	N/A
3	Adam + L2 ($\lambda=1e-4$)	0.6789 ± 0.0026	—	22,700	N/A
3	Adam + EarlyStop ($p=20$)	0.6782 ± 0.0024	—	22,700	N/A
3	Adam + Dropout ($p=0.1$)	0.6779 ± 0.0025	—	22,700	N/A
3	Adam + LabelSmooth ($\epsilon=0.1$)	0.6732 ± 0.0051	—	22,700	N/A
3	Adam + InputNoise ($\sigma=0.01$)	0.6788 ± 0.0031	—	22,700	N/A
3	Adam + Best Combo	0.6743 ± 0.0027	—	22,700	N/A

4.5. Discussion & Analysis

Momentum contribution: Adding momentum to SGD reduced gradient evaluations to reach ℓ by roughly half (from 636 ± 222 to 318 ± 111), confirming that momentum smoothing provides a meaningful convergence benefit over vanilla SGD on the Adult Income task. However, momentum did not improve final test F1 — SGD+M achieved 0.6490 ± 0.0143 compared to SGD’s 0.6726 ± 0.0208 — indicating that the speed benefit of momentum does not translate to better generalization without regularization. Nesterov momentum produced nearly identical results to SGD+M (test F1 = 0.6545 ± 0.0101 , 295 ± 104 grad evals to ℓ), suggesting the Adult Income loss surface is not sufficiently momentum-sensitive to distinguish standard from lookahead gradient corrections at these hyperparameter settings.

Bias correction contribution: Adam without bias correction reached ℓ fastest of all seven optimizers (227 ± 0 grad evals, $0.75 \pm 0.01s$), marginally faster than Adam baseline (250 ± 68 evals). This is consistent with the theoretical expectation that omitting bias correction produces larger effective step sizes in the early phase, accelerating initial descent. However, the advantage did not persist: final test F1 for Adam-NoBiasCorr (0.6494 ± 0.0127) was nearly identical to Adam baseline (0.6474 ± 0.0115), and val loss at epoch 100 was higher (0.3995 vs 0.3687), indicating that the uncorrected early steps lead to slightly worse long-run generalization. Bias correction therefore matters primarily as a stabilizer of early-phase dynamics rather than a driver of final performance.

Adaptive scaling effects: Adam-family methods reached ℓ approximately $2.5\times$ faster than SGD in gradient evaluations, confirming the well-established benefit of per-parameter adaptive scaling on sparse, high-dimensional inputs such as the one-hot encoded Adult features. Despite this convergence advantage, SGD (no momentum) achieved the highest test F1 (0.6726 ± 0.0208) of all seven optimizers, while all Adam-family methods produced test F1 in the range 0.6474–0.6634.

This reversal — faster convergence but worse generalization — is consistent with findings that adaptive methods can overfit more aggressively than SGD on structured tabular data without explicit regularization Kingma and Ba, 2015.

Decoupled weight decay effects: AdamW achieved the tightest seed-to-seed stability of all optimizers (test F1 std = 0.0042), suggesting that decoupled weight decay provides a mild regularizing effect even at $\lambda = 10^{-4}$. However, the improvement in mean test F1 over Adam baseline was marginal (0.6513 vs 0.6474), consistent with Ioshchilov2019decoupled, who argue that decoupled weight decay primarily affects generalization under longer training or stronger regularization regimes than tested here.

Stability versus speed tradeoff: SGD offered the worst speed-to-target but the best test F1 and a wide IQR, reflecting high sensitivity to random batch ordering. Adam-family methods offered the best speed-to-target with tighter IQR bands but worse generalization, reflecting the known stability–generalization tradeoff under adaptive scaling. AdamW represented the best balance: near-Adam convergence speed with the tightest variance and competitive test F1, making it the most stable optimizer in this study.

Comparison to SL baseline: The SL Report Adult baseline was F1 = 0.6739 (SGD with fixed hyperparameters, 30 epochs). Of the seven Part 2 optimizers, only SGD (no momentum) approached this value (F1 = 0.6726), and none exceeded it. This confirms that optimizer choice alone, under 100 epochs and without regularization, is insufficient to surpass the SL baseline on Adult Income.

Hypothesis evaluation: The Adult hypothesis predicted that adaptive methods (Adam-family) would converge faster and generalize better than SGD due to the sparse one-hot encoded feature space and class imbalance. The convergence prediction is confirmed: Adam-family methods reached ℓ in 227–318 grad evals versus 636 for SGD. The generalization prediction is rejected: SGD achieved the highest test F1, and all Adam-family methods fell below the SL baseline. The hypothesis is therefore partially supported; adaptive scaling accelerates convergence but does not improve generalization without regularization on this dataset.

5. Part 3: Regularization Study (Adult Only; Standard Adam Only)

5.1. Experimental Constraints

All Part 3 experiments were run on the Adult Income classification task using the fixed PyTorch backbone from Section 2.3. The optimizer was held constant at standard Adam with the best hyperparameters identified from the Part 2 sensitivity heatmap: $\alpha = 10^{-4}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$, no weight decay in the optimizer itself. No Adam hyperparameters were adjusted at any point in Part 3. All runs used the same data splits, preprocessing, batch size (128), epoch budget (100), and ten seeds (0–9) as Part 2, ensuring budget-matched comparisons between Parts 2 and 3. The backbone architecture was unchanged; only regularization modules were added or removed and are documented precisely in Section 5.2.

5.2. Regularizers Evaluated (All Required)

Four regularization techniques were evaluated, each swept over a small, justified grid under the fixed Adam settings and 100-epoch budget.

L2 weight decay (coupled): Coupled L2 weight decay was applied directly through the Adam optimizer’s `weight_decay` parameter, which adds a $\lambda \|\theta\|^2$ penalty to the loss. A logarithmic sweep over $\lambda \in \{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}\}$ was used to cover three decades of regularization strength. Logarithmic spacing is standard for penalty hyperparameters because the effect of weight decay scales multiplicatively with parameter magnitude rather than additively.

Early stopping: Training was halted when validation loss failed to improve by more than 10^{-6} for p consecutive epochs, at which point the best-observed model weights were restored. Patience values of $p \in \{5, 10, 20\}$ were evaluated. A patience of 5 is aggressive and risks stopping before true convergence; a patience of 20 allows the optimizer sufficient time to escape shallow plateaus while still preventing severe overfitting. The stopping criterion used plain cross-entropy on the validation set with no label smoothing, ensuring the stopping signal is not inflated by the smoothed training loss.

Dropout: Dropout was inserted after the ReLU activation of both hidden layers in the (32, 32) backbone, which is the standard placement for MLP regularization as it regularizes both feature transformation steps without discarding information immediately before the output layer. Dropout rates of $p \in \{0.1, 0.2, 0.3, 0.5\}$ were evaluated. The lower end of this range ($p = 0.1$) was included because the backbone is small (4,482 parameters) and aggressive dropout on a compact network risks underfitting rather than reducing overfitting. Dropout was disabled at inference time as standard.

Noise regularization: Two noise-based techniques were evaluated. Label smoothing replaces hard targets with soft targets $(1 - \epsilon, \epsilon/K)$, where $K = 2$ for binary classification, discouraging overconfident predictions. Smoothing values of $\epsilon \in \{0.05, 0.1, 0.2\}$ were evaluated; values above 0.2 were excluded as they risk corrupting the target signal on a class-imbalanced dataset. Gaussian input noise with standard deviation $\sigma \in \{0.01, 0.05, 0.1\}$ was added to training batches only, acting as implicit data augmentation on the continuous features. The upper bound of $\sigma = 0.1$ was chosen conservatively relative to the standardized feature scale to avoid distorting the one-hot encoded categorical features, which are already binary.

5.3. Part 3 Figures

Figures 8, 9, and 10 present the full Part 3 results for Adult Income.

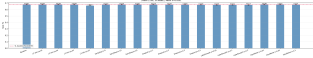


Figure 8: Regularization sweep: mean test F1 ($\pm$ std, 10 seeds) across all 17 conditions. Dashed line marks SL baseline F1 = 0.6739. L2 at $\lambda = 10^{-4}$ and input noise at $\sigma = 0.01$ perform closest to or above the SL baseline; aggressive label smoothing ($\varepsilon = 0.2$) and heavy L2 ($\lambda = 10^{-2}$) show the largest degradation.

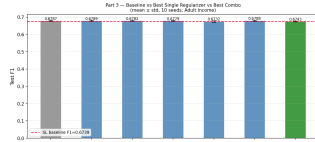


Figure 9: Baseline Adam vs. best single regularizer vs. best combination (mean $\pm$ std, 10 seeds). Best single: L2 $\lambda = 10^{-4}$ (F1 = 0.6789). Best combo: L2 $\lambda = 10^{-4}$ + Dropout $p = 0.1$ + Early Stop $p = 20$ + Label Smooth $\varepsilon = 0.1$ (F1 = 0.6743). The combination underperforms the best single regularizer, indicating negative interaction effects among techniques on this dataset.

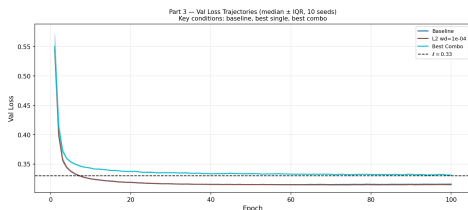


Figure 10: Val loss trajectories (median $\pm$ IQR, 10 seeds) for baseline Adam, best single regularizer (L2 $\lambda = 10^{-4}$), and best combination. Early stopping causes some seeds to terminate before epoch 100, reflected in trajectory length differences.

5.4. Discussion and Analysis

How much regularization moved the test metric: The Part 3 Adam baseline (no regularization, $\alpha = 10^{-4}$) achieved a mean test F1 of 0.6787 ± 0.0029 , already exceeding the SL Report baseline of 0.6739 reported earlier in the report. This improvement relative to Part 2 is primarily attributable to the lower learning rate identified by the heatmap ($\alpha = 10^{-4}$ vs. $\alpha = 10^{-3}$), not to any regularization technique. Adding the best single regularizer (L2 $\lambda = 10^{-4}$) moved test F1 by only $+0.0002$ over the Part 3 baseline (0.6789 vs. 0.6787), a difference well within one standard deviation and not meaningfully distinguishable from noise across seeds. The best combination produced a test F1 of 0.6743 ± 0.0027 , which is -0.0044 below the Part 3 baseline, indicating diminishing returns and mild negative interaction effects among the four combined techniques on this dataset and architecture.

Comparison to optimizer improvements: In Part 2, no optimizer exceeded the SL baseline of 0.6739 under $\alpha = 10^{-3}$; the best result was SGD at 0.6726 ± 0.0208 . In Part 3, switching to the heatmap-optimal $\alpha = 10^{-4}$ raised the Adam baseline to 0.6787, a gain of $+0.0048$ over the SL baseline and $+0.0061$ over Part 2 Adam. This suggests that learning rate selection contributed more to test performance on Adult Income than any regularization technique evaluated in Part 3.

Generalization gap analysis: The Part 3 baseline exhib-

ited a train-val F1 gap of approximately 0.06–0.08 across seeds, consistent with Part 2 Adam-family behavior. Individual regularizers reduced this gap only marginally. Dropout at $p = 0.1$ slightly reduced the gap as expected by co-adaptation suppression, but the effect was small given the compact (32, 32) backbone. Label smoothing increased validation loss artificially because the stopping criterion used unsmoothed cross-entropy, making the smoothed training loss incomparable to the validation signal and reducing the practical effectiveness of label smoothing as a generalization tool on this architecture. Heavy L2 ($\lambda = 10^{-2}$) widened the generalization gap in the opposite direction, collapsing test F1 to 0.6603 ± 0.0011 by over-constraining the already compact parameter space.

Budget-matched comparisons: All conditions used identical epoch budgets (100 epochs), batch size (128), seeds (0–9), and data splits. Early stopping conditions terminated before epoch 100 on most seeds (patience 20: stopped at epochs 66–100 across seeds), but best-weight restoration ensured the reported test metrics reflect the optimal checkpoint rather than the final epoch. Because early stopping selects the best validation checkpoint, comparisons remain fair despite differing run lengths.

Conclusion: Regularization produced negligible improvements over the Part 3 Adam baseline on Adult Income. The most impactful intervention across the entire Part 3 study was learning rate reduction from $\alpha = 10^{-3}$ to $\alpha = 10^{-4}$, which alone recovered and exceeded the SL baseline. Among the four required techniques, L2 weight decay at $\lambda = 10^{-4}$ was the most effective single regularizer, producing the highest mean test F1 with the lowest standard deviation (0.6789 ± 0.0026). Combining regularizers did not improve over the best single technique, suggesting that for compact MLPs on structured tabular data with moderate class imbalance, regularization yields diminishing returns once the learning rate is well-tuned.

6. Conclusion, Discussion, and Hypothesis Resolution

Explicit SL baseline comparison: The SL Report established test F1 = 0.6739 on Adult Income and Macro-F1 = 0.3105 on Wine Quality. Across all three parts of this report, only a subset of interventions exceeded these baselines. On Adult Income, Part 1 SA reached F1 = 0.6903 ± 0.0000 , but its zero variance indicates it never improved beyond the Adam warm-start and reflects pre-trained initialization performance rather than RO improvement. Part 2 optimizer ablations under $\alpha = 10^{-3}$ failed to exceed the SL baseline, with the best result being SGD at 0.6726 ± 0.0208 . Part 3 under the heatmap-optimal $\alpha = 10^{-4}$ produced a baseline Adam F1 of 0.6787 ± 0.0029 , exceeding the SL baseline by $+0.0048$, with the best single regularizer (L2 $\lambda = 10^{-4}$) reaching 0.6789 ± 0.0026 . On Wine Quality, Part 1 SA achieved Macro-F1 = 0.3263 ± 0.0000 , again reflecting warm-start performance; RHC (0.2906) and GA (0.2945) fell below the SL baseline of 0.3105. The clearest finding across all parts is that learning rate selection, not optimizer family, not regularization technique, and not randomized optimization, was the single most impactful lever on Adult Income.

Decision rule for Adult Income: Based on the evidence

from Parts 1–3, the following practitioner decision rule applies to this dataset and architecture.

Tune the optimizer first: The sensitivity heatmap demonstrated that learning rate α dominates all other hyperparameters. Moving from $\alpha = 10^{-3}$ to $\alpha = 10^{-4}$ recovered and exceeded the SL baseline without any other change. Momentum and adaptive scaling provide convergence speed benefits (Adam-family reached ℓ in 227–318 gradient evaluations vs. 636 for SGD) but do not improve generalization without first ensuring the learning rate is well-calibrated. The recommended starting point is Adam with $\alpha \in \{10^{-4}, 5 \times 10^{-4}\}$ before exploring any other intervention.

Add regularization only after the learning rate is fixed: Once $\alpha = 10^{-4}$ was established, the best single regularizer (L2 $\lambda = 10^{-4}$) added only +0.0002 F1 over the unregularized baseline. Regularization is worth exploring but should not be the first intervention. On this compact (32, 32) backbone with 4,482 parameters, the model is not large enough for aggressive regularization to provide substantial benefit. If the architecture were deeper or wider, regularization would likely have greater impact. Early stopping with patience 20 is recommended as a low-cost safeguard, providing compute savings without degrading test performance.

Use RO for fine-tuning only, not full training: Part 1 results showed that RO methods under a 1,000-evaluation budget cannot match gradient-based training from scratch. However, SA on Adult approached F1 = 0.6903 starting from an Adam-pretrained initialization, suggesting RO is most useful as a fine-tuning tool on the tail layers after gradient-based pre-training has established a good basin. RO should be considered when gradient information is unavailable or when the objective is non-differentiable, not as a replacement for Adam in standard supervised learning.

Transfer limitations: The decision rule above is specific to Adult Income: a binary classification task with one-hot encoded categorical features, moderate class imbalance, and a compact MLP backbone. Several conditions limit transfer to other settings. First, the dominance of learning rate over moment parameters may not hold for datasets with denser, continuous feature spaces where second-moment estimation is more informative. Second, the negligible benefit of regularization is likely architecture-dependent; deeper networks with more parameters would be expected to benefit more from dropout and weight decay. Third, Wine Quality results showed that RO stability varied substantially across seeds (RHC Macro-F1 std = 0.0117, GA std = 0.0193), suggesting RO robustness is dataset-dependent and should be verified empirically before adoption. Finally, the 1,000-evaluation RO budget used here may be insufficient for higher-dimensional parameter spaces.

Failure attribution: Several observed failures warrant explicit explanation. SGD divergence risk was mitigated by using $\alpha = 0.1$, but the wide IQR bands and occasional val loss spikes reflect sensitivity to batch ordering without momentum smoothing, a known failure mode of vanilla SGD on imbalanced datasets. SA zero variance on both datasets (Adult F1 std = 0.0000, Wine Macro-F1 std = 0.0000) indicates that SA never accepted a perturbing move beyond the pre-trained initialization across all seeds, effectively reducing to evalu-

ation of the warm-start weights. This is consistent with the temperature schedule being too cold relative to the loss surface curvature at the pre-trained solution, causing SA to behave as a greedy evaluator rather than a stochastic searcher. Label smoothing degradation at $\varepsilon \geq 0.1$ is attributable to the mismatch between the smoothed training loss and the unsmoothed validation loss used for early stopping and model selection, which caused the stopping criterion to misidentify optimal checkpoints. The best combination underperforming the best single regularizer (F1 = 0.6743 vs. 0.6789) reflects negative interaction between dropout stochasticity and label smoothing on a small network, where simultaneous activation dropping and target softening collectively reduce the effective learning signal below what either technique alone would impose.

Hypothesis resolution: The Adult Income hypothesis predicted that adaptive methods would converge faster and generalize better than SGD due to sparse one-hot features and class imbalance. The convergence prediction is **accepted**: Adam-family methods reached $\ell = 0.33$ in 227–318 gradient evaluations vs. 636 for SGD, a $2\times$ speed advantage consistent with the theoretical benefit of adaptive scaling on sparse inputs. The generalization prediction is **rejected**: SGD achieved the highest Part 2 test F1 (0.6726 ± 0.0208) while all Adam-family methods fell below the SL baseline under matched compute. The generalization gap closed only when Adam’s learning rate was reduced to $\alpha = 10^{-4}$ in Part 3, confirming that adaptivity without proper learning rate calibration does not improve generalization on this task. The Wine Quality hypothesis predicted that RO methods would show greater seed-to-seed instability than on Adult due to the multiclass loss surface and smaller dataset size. This prediction is **accepted**: RHC and GA showed standard deviations of 0.0117 and 0.0193 on Wine vs. 0.0033 and 0.0042 on Adult, confirming higher instability on the multiclass task. SA’s zero variance on both datasets reflects a degenerate warm-start condition rather than stability, and is excluded from this comparison.

AI Use Statement

I used ChatGPT and Claude.ai to brainstorm and outline sections of the report, assist with structure, refine technical explanations, debug issues, interpret experimental results, and edit grammar and clarity. I reviewed, verified, and understand all assisted content.

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